Data Collection and Preprocessing Phase

Date	28 Aug 2026
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Project Title	**Global AI Adoption in Education: A Comprehensive Global Performance Intelligence Report**
Maximum Marks	10 Marks

Data Exploration and Preprocessing

The **AI in Education dataset (2015–2026)** was explored and prepared before importing it into Tableau Public. The preprocessing stage focused on **understanding the available fields, checking data types, identifying missing or inconsistent values, and organizing the data** for effective visualization, dashboard creation, and Tableau Story development.

Section	Description
Data Overview	The dataset contains structured information related to **Artificial Intelligence adoption and usage in the education sector from 2015–2026**. The major fields include **Country, Year, Student AI Usage, Teacher AI Usage, School AI Adoption, AI Adoption Level, Community Type, Urban AI Usage, Rural AI Usage, and AI Usage Growth**. These attributes were used to analyze the development of AI in education, compare student and teacher usage across countries, examine yearly adoption trends, identify different adoption levels, and compare AI usage between urban and rural communities.

Data Cleaning	Missing Values: Important fields such as Country, Year, Student AI Usage, Teacher AI Usage, School AI Adoption, and AI Adoption Level were reviewed for blank or incomplete values before analysis. Duplicates: Dataset records were checked to identify and avoid duplicate entries that could affect the accuracy of usage and adoption analysis. Data Consistency: Categorical values such as Country, AI Adoption Level, and Community Type were reviewed for consistency. Inconsistent text formatting, unnecessary spaces, and variations in category names were standardized where required. Numerical fields were also checked to ensure that their values were suitable for analysis and visualization.
Data Transformation	Calculated Fields: Tableau calculations were used where required to calculate and analyze average AI usage, adoption percentages, usage growth, and other analytical measures. AI Adoption Classification: Records were categorized into adoption levels such as High, Moderate, and Low AI Adoption to understand the distribution of AI adoption in education. Aggregation: SUM, AVG, COUNT, and COUNTD measures were used to analyze student usage, teacher usage, school adoption, country-wise records, and adoption levels.

	Filtering: Filters were applied to fields such as Country, Year, AI Adoption Level and Community Type to support interactive analysis and allow users to explore specific parts of the dataset.
Data Type Conversion	Data fields were validated and assigned appropriate data types according to their usage in Tableau. **Country, AI Adoption Level, and Community Type** were treated as categorical dimensions, while **Student AI Usage, Teacher AI Usage, School AI Adoption, Urban AI Usage, Rural AI Usage, and AI Usage Growth** were treated as numerical measures. **Year** was treated as a time-related field and was used for analyzing changes in AI adoption and usage from 2015 to 2026. Appropriate data types were maintained to ensure accurate calculations, filtering, sorting, and visualization in Tableau.
Column Splitting and Merging	The dataset already contained separate fields for important education-related attributes such as **Country, Year, Student AI Usage, Teacher AI Usage, School AI Adoption, AI Adoption Level, and Community Type**. These fields were used independently where required and combined within Tableau visualizations to create meaningful comparisons. Related fields were analyzed together for visualizations such as **Country vs Student and Teacher AI Usage, Year vs AI Adoption, AI Adoption Level Distribution, and Urban vs Rural AI Usage**. This approach provided flexibility for creating multiple views from the same dataset.
Data Modeling	The dataset was organized around important dimensions such as **Country, Year, AI Adoption Level, and Community Type**. These dimensions were analyzed against numerical measures including **Student AI Usage, Teacher AI Usage, School AI Adoption, Urban AI Usage, Rural AI Usage, and AI Usage Growth**. Tableau aggregation functions such as **SUM, AVG, COUNT, and COUNTD** were used to generate analytical metrics and support KPI visualizations. The data structure allowed different dimensions and measures to be combined effectively for country-wise comparisons, time-series analysis, and community-level analysis.
Save Processed Data	After the data was reviewed, cleaned, and prepared, the structured dataset was connected to **Tableau Public** for further analysis and visualization. The processed data was used as the primary data source for creating **Tableau worksheets, calculated fields, filters, KPI visualizations, interactive dashboards, and the final Tableau Story**. The prepared dataset was retained in its structured tabular format so that it could be reused for additional analysis, visualization development, and modifications to the project.

Data Exploration Performed

During the exploration stage, the following important fields were examined:

- **Country** – To identify and compare AI education trends across different countries.
- **Year** – To analyze changes in AI usage and adoption from 2015 to 2026.
- **Student AI Usage** – To understand the level of AI usage among students.
- **Teacher AI Usage** – To examine AI usage patterns among teachers.
- **School AI Adoption** – To analyze the adoption of AI technologies across educational institutions.
- **AI Adoption Level** – To categorize records into **High, Moderate, and Low** AI adoption levels.
- **Community Type** – To distinguish between **Urban and Rural** communities.
- **Urban AI Usage** – To analyze AI usage and adoption patterns in urban communities.
- **Rural AI Usage** – To analyze AI usage and adoption patterns in rural communities.
- **AI Usage Growth** – To examine the growth and development of AI usage over the selected years.
- **Student vs Teacher AI Usage** – To compare AI usage levels between students and teachers across different countries.
- **School vs Individual Adoption** – To understand the relationship between institutional AI adoption and student/teacher usage.

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Preprocessing Outcome

After exploration, cleaning, and preparation, the dataset was found suitable for Tableau-based Business Intelligence analysis. The processed data supported the creation of the project's major visualizations, including:

1. Geographic Trends in Education Development
2. Global Comparison of Student and Teacher AI Usage
3. AI Adoption Journey in Education
4. AI Adoption Level Breakdown
5. Growth of AI Usage Across Urban and Rural Communities
6. Country-wise AI Usage Comparison
7. Key Performance Indicators (KPIs)

The processed dataset was then used to develop the project's interactive dashboards and Tableau Story, providing insights into global AI adoption, usage patterns, yearly growth, and differences between countries and communities in the education sector.